ChatGPT said:

AI Course-II AIML/REC – Category 1

Practical Lab Tasks – 24/07/2026

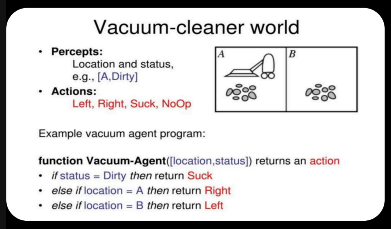
Category 1 focuses on implementing the four types of Intelligent Agents using Python.

I'll use simple examples with easy-to-understand code, sample outputs and explanations suitable for your REC practical examination.

Programs covered

No.	Intelligent Agent	Example
1	Simple Reflex Agent	Vacuum Cleaner
2	Model Based Agent	Vacuum Cleaner with Memory
3	Goal Based Agent	Robot Navigation
4	Utility Based Agent	Smart Route Selection

1. Simple Reflex Agent



Aim: To implement a Simple Reflex Agent using Python.

Concept:

A Simple Reflex Agent selects an action based only on the current condition (percept). It does not remember previous states.

Python Program

Python

```
# Simple Reflex Agent - Vacuum Cleaner

def vacuum_agent(location, status):

    if status == "Dirty":
        return "Suck"

    elif location == "A":
        return "Move Right"

    else:
        return "Move Left"


# Environment
rooms = {
    "A": "Dirty",
    "B": "Dirty"
}

location = "A"

for i in range(4):

    status = rooms[location]

    print("Location:", location)
    print("Status:", status)
```

Sample Output

Plain text

```
Location: A
Status: Dirty
Action: Suck

Location: A
Status: Clean
Action: Move Right

Location: B
Status: Dirty
Action: Suck

Location: B
Status: Clean
Action: Move Left

Final Environment: {'A': 'Clean', 'B': 'Clean'}
```

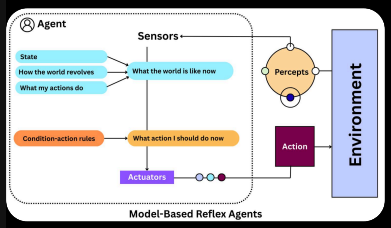
Explanation:

1. Create two rooms, A and B.
2. Assign their cleaning status.
3. The agent observes its current location and status.
4. If the room is dirty, it performs Suck.
5. If the room is clean, it moves to another room.
6. Repeat the process.

Key point: Simple Reflex Agent uses condition-action rules.

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2. Model Based Agent



Aim: To implement a Model Based Agent using Python.

Concept:

A Model Based Agent maintains an internal state (memory) of the environment. It uses previous information along with the current percept to make decisions.

Python Program

Python

```
# Model Based Agent - Vacuum Cleaner

class ModelBasedAgent:

    def __init__(self):
        self.model = {
            "A": "Unknown",
            "B": "Unknown"
        }

    def action(self, location, status):
```

```
        # Update internal model
        self.model[location] = status

        if status == "Dirty":
            return "Suck"

        elif "Unknown" in self.model.values():
            if self.model["A"] == "Unknown":
                return "Move A"
            else:
                return "Move B"

        else:
            return "Stop"
```

Sample Output

Plain text

```
Location: A
Status: Dirty
Action: Suck
Internal Model: {'A': 'Dirty', 'B': 'Unknown'}

Location: A
Status: Clean
Action: Move B
Internal Model: {'A': 'Clean', 'B': 'Unknown'}

Location: B
Status: Clean
Action: Stop
Internal Model: {'A': 'Clean', 'B': 'Clean'}
```

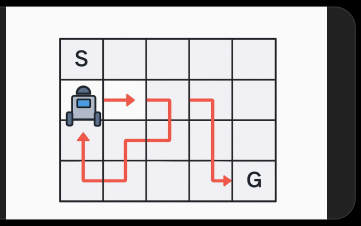
Explanation:

1. Create an internal model for rooms A and B.
2. Initially, both rooms are unknown.
3. Observe the current room.
4. Update the internal model.
5. If the room is dirty, clean it.
6. If another room is unknown, move there.
7. Stop when all rooms are known and clean.

Key point: Model Based Agent uses memory and an internal representation of the environment.

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3. Goal Based Agent



Aim: To implement a Goal Based Agent using Python.

Concept:

A Goal Based Agent selects actions based on a desired goal. It searches for a path to reach the goal.

Python Program

Python

```python
# Goal Based Agent - Robot Navigation

from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D"],
    "C": ["E"],
    "D": ["F"],
    "E": ["F"],
    "F": []
}

def goal_based_agent(start, goal):

    queue = deque([[start]])
    visited = set()

    while queue:

        path = queue.popleft()
        current = path[-1]

        if current == goal:
            return path

        if current not in visited:

            visited.add(current)
```

Sample Output

Plain text

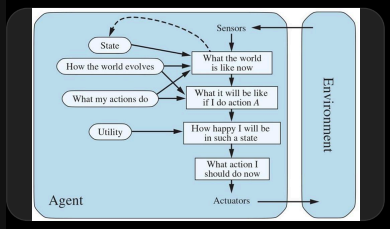
```
Goal Reached!
Path: ['A', 'C', 'E', 'F']
```

Explanation:

1. Create a graph representing different locations.
2. Define the starting position and goal.
3. Insert the starting position into the queue.
4. Explore the neighboring locations.
5. Check whether the current location is the goal.
6. If the goal is reached, display the path.

Key point: Goal Based Agent works towards achieving a specified goal.

4. Utility Based Agent



Aim: To implement a Utility Based Agent using Python.

Concept:

A Utility Based Agent evaluates different possible actions and selects the one with the highest utility value.

For example, a delivery robot chooses a route based on travel time.

Python Program

Python

```python
# Utility Based Agent - Route Selection

routes = {
    "Route A": 30,
    "Route B": 20,
    "Route C": 40
}

def utility_agent(routes):

    best_route = None
    best_utility = float("-inf")

    for route, time in routes.items():

        utility = 100 - time

        print(route, "Time:", time,
              "Utility:", utility)

        if utility > best_utility:
            best_utility = utility
            best_route = route

    return best_route


result = utility_agent(routes)
```

Sample Output

Plain text

```
Route A Time: 30 Utility: 70
Route B Time: 20 Utility: 80
Route C Time: 40 Utility: 60

Best Route: Route B
```

Explanation:

1. Create different routes with their travel times.
2. Calculate the utility of each route.
3. Higher utility represents a better route.
4. Compare all utility values.
5. Select the route with the highest utility.

Key point: Utility Based Agent selects the action that maximizes utility.

Important viva questions

Question	Answer
What is an intelligent agent?	An entity that perceives its environment and performs actions.
What is a Simple Reflex Agent?	An agent that acts based on the current percept.
What is a Model Based Agent?	An agent that maintains an internal state of the environment.
What is a Goal Based Agent?	An agent that selects actions to achieve a goal.
What is a Utility Based Agent?	An agent that selects actions based on utility values.
What is a percept?	Information received from the environment through sensors.

What is an actuator?	A component used by an agent to perform actions.
What is an internal state?	Information stored by an agent about the environment.
What is a goal?	A desired state that an agent tries to achieve.
What is utility?	A numerical measure of how desirable an outcome is.

Comparison of four Intelligent Agents

Feature	Simple Reflex	Model Based	Goal Based	Utility Based
Uses current percept	Yes	Yes	Yes	Yes
Uses memory	No	Yes	Can use	Can use
Has goal	No	No	Yes	Can have
Evaluates utility	No	No	No	Yes
Decision basis	Rules	Internal model	Goal	Highest utility

Final practical revision – Categories 1 to 5

You now have programs covering all five categories:

1. **Category 1:** Intelligent Agents – 4 programs.
2. **Category 2:** Environment Types – 6 programs.
3. **Category 3:** Local Search – 3 programs.
4. **Category 4:** Uninformed Search – 4 programs.
5. **Category 5:** Heuristic Search – 4 programs.

21 practical implementations covered

My suggestion: For your practical exam, understand the four agent programs first. Category 1 is the foundation for Category 2 because Category 2 asks you to implement agents under different environmental conditions.